

Library Management System in Z Notation

Introduction

This document defines a Library Management System using Z notation. The system manages books and users in a library. It includes two types of users: **STUDENT** and **TEACHER**, with different rules for borrowing.

The system is divided into:

Static Part:

- Describes the state of the library.
- Includes available books, borrowed books, user types, borrowing dates, and today's date.
- Maintains rules like borrowed books must come from the available set.

Operational Part:

- Borrowing a book
- Returning a book
- Setting due dates
- Calculating fines

Basic Types

$$[BOOK, USER, DATE] USER_TYPE ::= STUDENT \mid TEACHER$$

State Schema

Library

$available : \mathbb{P} BOOK$
 $borrowed : BOOK \rightarrow USER$
 $userType : USER \rightarrow USER_TYPE$
 $borrowDate : BOOK \rightarrow DATE$
 $today : DATE$

$\text{dom } borrowed \subseteq available$
 $\text{dom } borrowDate = \text{dom } borrowed$

Constants and Rules

$MaxBorrow : USER_TYPE \rightarrow \mathbb{N}$
 $MaxBorrow = \{STUDENT \mapsto 3, TEACHER \mapsto 5\}$

Operations

$BorrowBook$ $\Delta Library$ $b? : BOOK$ $u? : USER$
$b? \in available$ $b? \notin \text{dom } borrowed$ $(\#\{b : BOOK \mid borrowed\ b = u?\}) < MaxBorrow\ (userType\ u?)$ $borrowed' = borrowed \cup \{b? \mapsto u?\}$ $borrowDate' = borrowDate \cup \{b? \mapsto today\}$ $available' = available$ $userType' = userType$ $today' = today$

$ReturnBook$ $\Delta Library$ $b? : BOOK$
$b? \in \text{dom } borrowed$ $borrowed' = \{b : BOOK \mid b \neq b?\} \triangleleft borrowed$ $borrowDate' = \{b : BOOK \mid b \neq b?\} \triangleleft borrowDate$ $available' = available$ $userType' = userType$ $today' = today$

$CalculateFine$ $\Xi Library$ $b? : BOOK$ $fine! : \mathbb{N}$
$b? \in \text{dom } borrowDate$ $let\ due : DATE == borrowDate\ b? + (if\ userType\ (borrowed\ b?) = STUDENT\ then\ 7\ else\ 14)$ $in\ fine! = if\ today > due\ then\ (today - due) * 10\ else\ 0$